ASTRODYNAMICS HW#1 SOLUTIONS (Spring 2017)

1. A satellite is in a circular parking orbit about the Earth (120km altitude). To begin a voyage to Moon, what velocity does it need to escape from the Earth?
   Would your answer be different for an elliptical orbit ($e = 0.3$, $\nu = 145°$, a the same)? Why or why not?
   If you assume both burns are tangent to the existing velocity, what are the required increases in velocity for each case?

Sol)

a) For the circular parking orbit

$$r=6378.137\text{km}+120\text{km}=6498.137\text{km}$$

from vis-viva eq. (for circular orbit, a=r)

$$v_{circle} = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} = \sqrt{\frac{\mu}{r}} = \sqrt{\frac{3.986 \times 10^5 km^3/s^2}{6498.137km}} = 7.832027713km/s$$

The satellite should have at least velocity of a parabolic orbit in order to escape from the Earth. (for parabolic orbit, a=∞)

$$v_{parabola} = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} = \sqrt{\frac{2\mu}{r}} = \sqrt{\frac{2 \times 3.986 \times 10^5 km^3/s^2}{6498.137km}} = 11.07615981km/s = v_{escape}$$

The velocity needed to escape from the Earth if

$$\Delta\text{v} = v_{escape} - v_{circular} = 11.07615981km/s - 7.832027713km/s = 3.244132097k/s$$

b) For the elliptical orbit(e=0.3, ν=145°, a the same)

$$a = 6378.137\text{km} + 120\text{km} = 6498.137\text{km}$$

$$\text{r} = \frac{a(1-e^2)}{1+ecosv} = \frac{6498.137km(1-0.3^2)}{1+0.3 \times \cos(145°)} = 7839.934078km$$

$$v_{elliptic} = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} = \sqrt{3.986 \times 10^5 km^3/s^2 \left(\frac{2}{7839.934078km} - \frac{1}{6498.137km}\right)} = 6.351682484km/s$$

The escape velocity can be obtained from velocity of parabolic orbit

$$v_{escape} = v_{parabola} = \sqrt{\frac{2\mu}{r}} = \sqrt{\frac{2 \times 3.986 \times 10^5 km^3/s^2}{7839.934078km}} = 10.08387468km/s$$

The velocity needed to escape from the Earth is

$$\Delta v_{ellipse} = v_{escape} - v_{ellipse} = 10.08387468 km/s - 6.351682484\mathrm{km/s} = 3.732192192\mathrm{km/s}$$

The answer is different from previous one, although the semi-major axes in both cases are the same. This is because 'r' in both cases are different.

c) The same as b)

2. Is there a common point at which the velocity of a satellite in elliptical orbit equals the velocity in a circular orbit? If so, where?

Sol)

From vis-viva eq.

$$\mathrm{v} = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)}$$

for circular orbit (a=r)

$$v_{circle} = \sqrt{\frac{\mu}{r}}$$

For elliptical orbit

$$v_{elliptic} = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}}$$

$$v_{circle} = v_{elliptic} \Rightarrow \sqrt{\frac{\mu}{r}} = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a}} \Rightarrow r = a$$

$$\mathrm{r} = \frac{a(1-e^2)}{1+ecosv} \Rightarrow r = \frac{r(1-e^2)}{1+ecosv}$$

$$1 - e^2 = 1 + ecosv$$

$$\cos\mathrm{v} = -\mathrm{e}$$

$$\mathrm{v} = cos^{-1}(-e)$$

cf) for circular orbit (a=r)

ASTRODYNAMICS HW#1 SOLUTIONS (Spring 2017)

$$v_{circle} = \sqrt{\frac{\mu}{a_C}}$$

For elliptical orbit

$$v_{elliptic} = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a_E}}$$

$$\sqrt{\frac{\mu}{a_C}} = \sqrt{\frac{2\mu}{r} - \frac{\mu}{a_E}}$$

Assume that $\mathrm{a_C = a_E}$

$$\frac{2\mu}{\mathrm{a_E}} = \frac{2\mu}{r}$$

r should same as $\mathrm{a_E}$

a) yes

b) at $\nu = cos^{-1}(-e)$

3. Identify the type of orbits given the following data:
   A. $\vec{r} = 6872.0\hat{I} + 6872.0\hat{J}\ km,\ \vec{v} = -4.7028\hat{I} + 4.7028\hat{J}\ km/sec$
   B. $\vec{e} = 0.08\hat{I} + 0.45\hat{J} + 0.021\hat{K}$
   C. $\vec{h} = 1.6\hat{J}\ \mathrm{ER^2/TU}$
   D. $\vec{h} = -3.5\hat{K}\ \mathrm{ER^2/TU}$

Sol)

$A.\ \vec{r} = 6872.0\hat{I} + 6872.0\hat{J}\ km$

$\vec{v} = -4.7028\hat{I} + 4.7028\hat{J}\ km/sec$

$$\mathrm{r} = |\vec{r}| = \sqrt{6872.0km^2 + 6872.0km^2} = 9718.475601km$$

$$\mathrm{v} = |\vec{v}| = \sqrt{(-4.7028km/s)^2 + 4.7028km/s^2} = 6.650763541km/s$$

$$\mathrm{v} = \sqrt{\mu\left(\frac{2}{r} - \frac{1}{a}\right)} \Rightarrow a = \left(\frac{2}{r} - \frac{v^2}{\mu}\right)^{-1} = 10545.902593km$$

ASTRODYNAMICS HW#1 SOLUTIONS (Spring 2017)

$$\vec{h} = \vec{r} \times \vec{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 6872.0km & 6872.0km & 0 \\ -4.7028km/s & 4.7028km/s & 0 \end{vmatrix} = 64635.2832\hat{k}\ km^2/s$$

$$\mathrm{h} = \sqrt{\mu p} = \sqrt{\mu a(1-e^2)} \Rightarrow e = \sqrt{1 - \frac{h^2}{\mu a}} = \sqrt{1 - \frac{(64635.2832\hat{k}\ km^2/s)^2}{3.986 \times 10^5 km^3/s^2 \times 10545.902593km}} = 0.07846$$

$$\text{Elliptic orbit for } 0 < \mathrm{e} < 1$$

$B.\ \vec{e} = 0.08\hat{I} + 0.45\hat{J} + 0.021\hat{K}$

$$\mathrm{e} = |\vec{e}| = \sqrt{0.08^2 + 0.45^2 + 0.021^2} = 0.4575$$

$$\text{Elliptic orbit for } 0 < \mathrm{e} < 1$$

C. $\vec{h} = 1.6\hat{J}\ \mathrm{ER^2/TU}$

$$\mathrm{h} = 1.6ER^2/TU$$

$$\mathrm{h} = \sqrt{\mu p} = \sqrt{\mu a(1-e^2)} \Rightarrow e^2 = 1 - \frac{h^2}{\mu a}$$

$$\mathrm{r_p} = a(1-e) = \frac{p}{1+e} = \frac{h^2}{\mu(1+e)} > 1\mathrm{ER}$$

$$\mathrm{e} = 0:\ \text{Circular orbit where a} = 2.56\mathrm{ER}$$

$$0 < \mathrm{e} < 1:\ \text{Elliptical orbit whrer a} > 2.56\mathrm{ER}\ \text{(not infinity)}$$

$$\mathrm{e} = 1:\ \text{Parabola orbit where a} = \infty$$

$$1 < \mathrm{e} < 1.56:\ \text{Hyperbola orbit where a} < -1.786\mathrm{ER}$$

D. $\vec{h} = -3.5\hat{K}\ \mathrm{ER^2/TU}$

$$\mathrm{h} = 3.5ER^2/TU$$

$$\mathrm{e} = 0:\ \text{Circular orbit where a} = 12.25\mathrm{ER}$$

$$0 < \mathrm{e} < 1:\ \text{Elliptical orbit whrer a} > 12.25\mathrm{ER}\ \text{(not infinity)}$$

$$\mathrm{e} = 1:\ \text{Parabola orbit where a} = \infty$$

$$1 < \mathrm{e} < 11.25:\ \text{Hyperbola orbit where a} < -0.098\mathrm{ER}$$

4. Homes comet moves in an elliptical orbit of eccentricity 0.433 and perihelion distance 2.053 AU. Compare its velocities, both linear and angular, at perihelion and aphelion.

ASTRODYNAMICS HW#1 SOLUTIONS (Spring 2017)

Sol)

$$1\text{AU} = 1.4960\text{e}08\text{km}$$

$$1\mu_{\odot} = 1\text{G}M_{\odot} = 1.3274\text{e}11 km^3/s^2$$

$$\text{e} = 0.433$$

$$r_p = 2.053AU\ (3.0713e08km)$$

$$r_p = a(1-e)\ \Rightarrow a = \frac{r_p}{1-e} = 3.608\text{AU}\ (5.4167\text{e}8\text{km})$$

$$r_a = 2a - r_p = 5.1886AU\ (7.7622e8km)$$

$$\text{p} = \text{a}(1-e^2) = 2.9419AU\ (4.4012e8km)$$

From "Conservative of angular momentum"

$$\text{h} = r_a v_a = r_p v_p = {r_p}^2 \dot{v_p} = {r_a}^2 \dot{v_a}$$

$$\text{h} = \sqrt{\mu p} = 7.6434\text{e}9 km^2/s$$

Linear

$$\text{h} = r_p v_p\ \Rightarrow\ v_p = \frac{h}{r_p} = 24.8865 km/s$$

$$\text{h} = r_a v_a\ \Rightarrow\ v_a = \frac{h}{r_a} = 9.8469 km/s$$

Angular

$$\text{h} = {r_p}^2 \dot{v_p} \Rightarrow\ \dot{v_p} = \frac{\text{h}}{{r_p}^2} = 8.1030e-08 rad/s$$

$$\text{h} = {r_a}^2 \dot{v_a} \Rightarrow\ \dot{v_a} = \frac{\text{h}}{{r_a}^2} = 1.2686e-08 rad/s$$

In both case, linear and angular velocities at perihelion are greater than those at aphelion.

5. Consider a satellite in a "nearly circular" orbit about the Earth.

   A. Plot the gravitational acceleration of the satellite due to Earth's gravity as a function of the altitude of the satellite as measured from the surface of the Earth. Scale your plot so that the altitude goes from 0 km to 10,000 km.

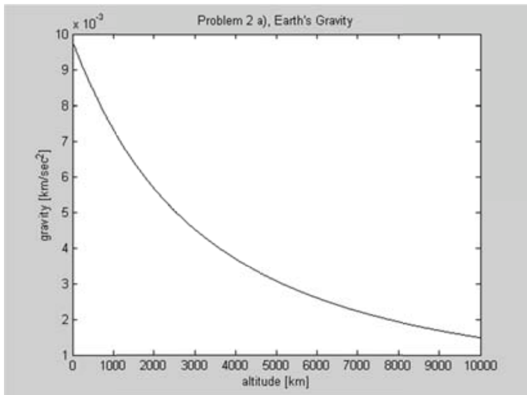
ASTRODYNAMICS HW#1 SOLUTIONS (Spring 2017)

B. Assume a satellite lies on the line between the Earth and the Sun. Make two plots of the gravitational acceleration of the satellite due to Earth's gravity and of the gravitational acceleration of the satellite due to the Sun's gravity, each as a function of the altitude of the satellite as measured from the surface of the Earth. Scale your plot so that the satellite altitude goes from 0 km to 10 million km.

C. Using your plots in part (b), assess the relative importance of Earth's gravity and Sun's gravity on the satellite's orbit. For what satellite altitudes do you think it is a reasonable approximation to ignore the Sun's gravity and use a two-body model for the satellite's orbit based only on Earth's gravity.

Sol)

a. Earth's gravity is given as $g=\frac{\mu}{(R_{\oplus}+h)^2}$ where h is the altitude.

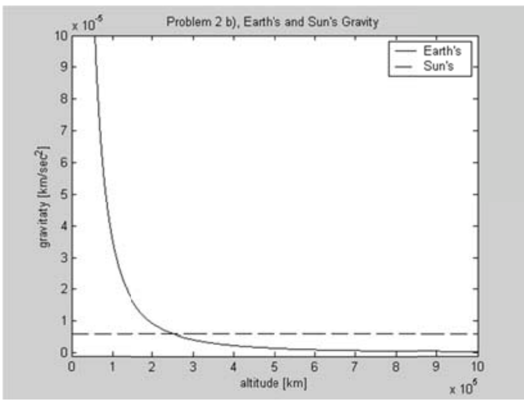


$$\frac{gM_{\oplus}m}{(R_{\oplus}+h)^2}=mg$$

$$g=\frac{\mu}{(R_{\oplus}+h)^2}$$

b. Sun's gravity is given as $g=\frac{\mu_{\odot}}{(R-R_{\oplus}-h)^2}$.

R : distance between Sun and Earth



$$mg_{\odot}=\frac{GM_{\odot}m}{(R-(R_{\oplus}+h))^2}$$

$$g=\frac{\mu_{\odot}}{(R-R_{\oplus}-h)^2}$$

c. As we can see in the figure, the Earth's gravity dominates the Sun's gravity when $h<2\times10^5$ km .

6. At a specific instant of time, an orbiting spacecraft is at an altitude of 1500 km above the Earth's surface; at this instant the velocity vector of the spacecraft has magnitude 7.8 km/ sec and flight path angle of 4.5 degrees.

a. What are the values of the local vertical velocity component $\dot{r}$ and the local horizontal velocity component r& at this instant?

b. What is the specific energy of this orbit?

c. What is the specific angular momentum of this orbit?

d. What type of conic section is the orbit?

Sol)

a. We know $\vec{v} = vsin\emptyset\overrightarrow{e_r} + vcos\emptyset$, $\overrightarrow{e_\theta} = \dot{r}\overrightarrow{e_r} + r\dot{\theta}\overrightarrow{e_\theta}$, where $\overrightarrow{e_r}$ and $\overrightarrow{e_\theta}$ are local horizontal and vertical unit vectors, respectively.

$$\mathrm{v_r} = v\sin\phi = 0.6120 km/s$$

$$\mathrm{v_\theta} = \mathrm{v}\sin\phi = 7.7760\mathrm{km/s}$$

b. The specific total energy of the orbit is given by

$$\mathrm{E} = \frac{1}{2}v^2 - \frac{\mu}{r} = -20.1757\ km^2/s^2$$

c. The specific angular momentum is given by

$$\mathrm{h} = \mathrm{rvcos}\emptyset = 6.13 \times 10^4 km^2/sec$$

d. Since the specific total energy is negative, the orbit is elliptical.

7. The Energy and Angular Momentum of several asteroids heading towards Earth have been measured. Using the Energy and Angular Momentum integrals only, determine which of these asteroids are a real problem, i.e., which ones will hit the Earth?

Hint: The angular momentum when closest to the Earth is H = r_p V_p, where r_p is the closest radius to the Earth, and V_p is the speed when closest to the Earth.

ASTRODYNAMICS HW#1 SOLUTIONS (Spring 2017)

Asteroid #	Energy (km^2/s^2)	Angular Momentum (km^2/s)
1	1	$1x10^5$
2	100	$1x10^5$
3	0	$7x10^4$
4	0	$8x10^4$
5	10	$8x10^4$

Asteroid #1

$$\mathrm{E} = -\frac{\mu}{2a} \rightarrow a = -\frac{\mu}{2E} = -\frac{3.986e5km^3/s^2}{2 \times 1km^2/s^2} = -1.9930\mathrm{e}5\mathrm{km}$$

$$\mathrm{h} = |\vec{r} \times \vec{v}| = \sqrt{\mu p} = \sqrt{\mu \cdot a(1-e^2)} \rightarrow \mathrm{e} = \sqrt{1 - \frac{h^2}{a\mu}} = \sqrt{1 - \frac{(1e5km^2/s)^2}{(-1.9930e5km)(3.986e5km^3/s^2)}} = 1.0611$$

$$r_p = a(1-e) = (-19930e5km) \cdot (1 - 1.0611) = 12177.23km \ > R_\oplus$$

There's nothing dangerous.

Asteroid #2.

$$a = -\frac{\mu}{2E} = -\frac{3.986e5km^3/s^2}{2 \times 100km^2/s^2} = -1993\mathrm{km}$$

$$\mathrm{e} = \sqrt{1 - \frac{h^2}{a\mu}} = \sqrt{1 - \frac{(1e5km^2/s)^2}{(-1993km)(3.986e5km^3/s^2)}} = 3.6862$$

$$r_p = a(1-e) = (-19930e5km) \cdot (1 - 3.6862) = 5353.5966km < R_\oplus$$

For $r_p < R_\oplus$, the asteroid #2 will collide with the Earth

Asteroid #3.

$\mathrm{a} = \infty, \mathrm{e} = 1$ from the E=0.

$$\mathrm{h} = |\vec{r} \times \vec{v}| = \sqrt{\mu p} = \sqrt{\mu \cdot r_p(1+e)} \ \rightarrow \ r_p = \frac{h^2}{\mu(1+e)} = \frac{(1e5km^2/s)^2}{(3.986e5km^3/s^2)(1+1)} = 6146.5128km < R_\oplus$$

For $r_p < R_\oplus$, the asteroid #3 will collide with the Earth

ASTRODYNAMICS HW#1 SOLUTIONS (Spring 2017)

Asteroid #4

$a = \infty, e = 1$ from the E=0.

$$r_p = \frac{h^2}{\mu(1+e)} = \frac{(8e4km^2/s)^2}{(3.986e5km^3/s^2)(1+1)} = 8028.0983km > R_\oplus$$

There's nothing dangerous.

Asteroid #5.

$$a = -\frac{\mu}{2E} = -\frac{3.986e5km^3/s^2}{2 \times 10km^2/s^2} = -19930\text{km}$$

$$\text{e} = \sqrt{1 - \frac{h^2}{a\mu}} = \sqrt{1 - \frac{(8e4km^2/s)^2}{(-19930km)(3.986e5km^3/s^2)}} = 1.34374$$

$$r_p = a(1-e) = (-19930e5km) \cdot (1 - 1.34374) = 6850.738km > R_\oplus$$

There's nothing dangerous.